

CV - Rachad Lakis

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Professional Summary

ML engineer with dual Master's degrees and published IEEE research, specializing in PyTorch, object detection, LLM fine-tuning, and distributed training (DDP/FSDP). I've built systems from scratch – from raw NumPy backpropagation to multi-GPU training pipelines – and have worked across the full AI stack: computer vision, agentic systems, RAG pipelines, chatbots, and MCP integrations.

EDUCATION

University Saint Joseph of Beirut (USJ) – Science Faculty & ESIB

2021 – 2023 | Master's in Data Science (Average: 16.5/20)

- Thesis: *Pose Estimation Key Points in Age Recognition of Full Body Images* (Published paper, Grade: 18/20)
- Publication: IEEE 4th International Multidisciplinary Conference on Engineering Technology (IMCET), 2023
- Main Courses: Machine Learning, Inferential Statistics, Graph Theory, Regression & Time Series Analysis, AI Fundamentals, Cloud & Digital Transformation, Big Data, Data Visualization, etc.

Lebanese International University (LIU) – Faculty of Science

2016 – 2018 | Master's in Computer Science (GPA: 3.8/4.0)

- Thesis: *Improved TFIDF Weighting Techniques in Document Retrieval* (Published, Grade: A)
- Publication: 13th International Conference on Digital Information Management (ICDIM), 2018
- Main Courses: Information Retrieval, Advanced Databases, Design & Analysis of Algorithms, Robotics, etc.

Lebanese International University (LIU) – Faculty of Science

2008 – 2013 | BS in Computer Science

- Thesis: *Major Change System for University internal usage*
- Main Courses: Data Structure and Algorithms Databases, Networking, Calculus, Object Oriented Programming, etc.

Lebanese University – Faculty of Science - Hadath

2006– 2008 | Pure Math (2 years)

- Main Courses: Algebra, Metric Topology, Vector Analysis and Multiple Integrals (curves, integral), Introduction to Probability Calculation, Linear Algebra, Sequences and Series of Functions, etc.
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WORK EXPERIENCE

devtech.pro – AI Consultant / Data Scientist (June 2023 – Present)

- Delivered multiple machine learning projects in computer vision, including image classification, object detection, and motion detection using Python and PyTorch.
- Designed, built, and deployed AI-powered chatbots using Large Language Models (LLMs), integrating Retrieval-Augmented Generation (RAG) pipelines with chunking, embedding, and storage in Qdrant vector databases.
- Implemented function calling and MCP integrations, enabling chatbots to interact with external APIs, modify databases, and automate workflows.

- Led end-to-end data science workflows: data collection, cleaning, preprocessing, analysis, and visualization.
- Developed AI-driven automation for content creation, social media publishing, automated email responses, and marketing workflows using low-code/no-code platforms such as n8n.

Prisma – Ideation & AI Developer (Feb 2025 - Present)

- Conceptualized and designed the idea of a chatbot as a service. Worked in the back end to add the reading, chunking, embedding and saving of texts to vector stores (PostGres). Then retrieving the most relevant ones to add data to user queries. Working on adding tools and MCP also.

UnifyApps – DGE / AI Architect & Consultant (Dec 2025 – May 2026)

- Consulted AI engineering teams on the design and implementation of large-scale AI workflows leveraging big datasets, agentic systems, RAG pipelines, and automation.
- Designed AI agent architectures to support enterprise and government use cases.
- Developed a comprehensive AI curriculum for DGE employees in the UAE (with focus on Abu Dhabi), covering AI fundamentals through advanced, production-level AI applications for government operations and daily workflows.

Argus – Machine Learning Engineer (Dec 2025 – Apr 2026)

- Built and curated a large-scale computer vision dataset of 120,000+ images for fire and smoke detection.
- Trained and evaluated multiple YOLO variants (n, s, m, l, xl) and RF-DETR models for real-time fire and smoke detection systems.
- Developed object detection models for firearm detection to enhance automated threat identification capabilities.

MightClick – Ideation & AI Developer (Jul 2024 – Oct 2024)

- Conceptualized and designed AI components for an AI-powered Google Chrome browser extension.
- Enabled users to generate text, comments, translations, summaries, audio, image understanding, and AI-generated content via a simple right-click interface.
- Collaborated on product ideation and AI feature design for a public-facing release (upcoming).

Robots Go Mental – Machine Learning Engineer (2024, 6-month project)

- Implemented Guided Transfer Learning (GTL) to improve neural network training efficiency with limited data.
- Fine-tuned Large Language Models (LLaMA 3.2-1B, BERT) using pure PyTorch without specialized LLM fine-tuning libraries.
- Conducted comparative experiments to evaluate model performance with and without GTL.
- Utilized AWS SageMaker for large-scale training and Amazon S3 for experiment tracking and result storage.

OMNIPeak – AI Solution Architect (Mar 2024 – Jun 2024)

- Designed AI-driven agentic workflows for B2B applications, integrating LLMs with APIs and CRM platforms, including HubSpot.
- Rapidly prototyped solutions using low-code platforms (n8n, Node-RED), followed by production implementation in pure Python.

AI Directions – Machine Learning Engineer (Master's Internship) (Jun 2023 – Sep 2023)

- Collected, trained, and evaluated machine learning models for age detection using transfer learning and fine-tuning techniques.
- Worked with CNN architectures including VGG-19, MobileNet-V2, and ResNet-50 for facial and non-facial age estimation.

Lebanese International University – Teaching Assistant (2017 – 2018)

- Assisted in teaching Intermediate Java and Object-Oriented Programming courses during Master's studies.
- Graded assignments and supported students with programming concepts and debugging.

AMHHS – Technical Consultant & System Administrator (2016 – 2021)

- Provided technical consulting, system administration, and IT infrastructure support.
 - Managed systems, resolved technical issues, and supported organizational IT operations.
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TECHNICAL SKILLS

Programming & Software Development

- Languages: Python (ML, data pipelines, deep learning), Java, C++, etc
- Hands-on Projects: Built AI and ML models both with and without frameworks (e.g., MNIST digit recognition from scratch using only NumPy).
- Tools & Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face, OpenCV.
- Development Practices: Asynchronous programming (Asyncio), package management with uv, pip, and conda.

Machine Learning & Artificial Intelligence

- Deep Learning: CNNs (ResNet, EfficientNet, MobileNet), Transformers, Guided Transfer Learning (GTL), and fine-tuning LLMs (BERT, LLaMA).
- Distributed Training: ZeRo, DDP, FSDP, etc.
- NLP: Prompt engineering, Retrieval-Augmented Generation (RAG), tokenization, and sentiment analysis.
- Model Optimization: Hyperparameter tuning, training on distributed architectures (DP, DDP, early FSDP).
- Frameworks: PyTorch, TensorFlow, Ultralytics, Hugging Face Transformers.

Data Science & Analytics

- Statistical Analysis: Regression, inferential statistics, time series forecasting, graph theory.
 - Dataset Engineering: Curated and normalized large-scale datasets (e.g., 120K+ image datasets merging COCO and YOLO formats).
 - Visualization: Matplotlib, Seaborn, Power BI, and custom dashboards for reporting insights.
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PROJECTS

- **OMNI-Train:**
Built a multi-GPU training framework (CLI and web app) in PyTorch to train/fine-tune with distributed training strategies (DDP and FSDP) for any model. Implemented DDP and FSDP – enabling making the training faster on GPUs and training of models that exceed single-GPU memory. Applied advanced memory optimization techniques including mixed precision (bfloat16), gradient checkpointing, and explicit layer prefetching to maximize throughput. Also OMNI-Train can support PEFT and quantization techniques. (Not FSDP and quantization, this is not possible for now). Based mainly on PyTorch and HuggingFace.
- **Agentic-ai framework:**
Designed and built a full-stack agentic AI platform from scratch using FastAPI, PostgreSQL, and PydanticAI. The framework supports multi-agent orchestration, tool calling, RAG (Retrieval-Augmented Generation), MCP (Model Context Protocol) integration, task scheduling via Celery, and real-time voice chat (STT/TTS).
- **Model Context Protocol – Sep 2025:**
Implemented Model Context Protocol (MCP) servers and clients to Cooriroo, a fleet management system. Fully wrote the MCP client and MCP server in Python.
- **Threat detection – Apr 2025:**
Object Detection Project to detect threats with a YOLO-v12 and RF-DETR object detection model, creating a

dataset of more than 120K images from different classes, by merging and normalizing datasets, training using Ultralytics and Roboflow frameworks based on PyTorch. Trying YOLOv12 nano, small, and others with different hyperparameters to get best results, and RF-DETR transformer based Object Detector to attain best mAP and precision.

- **Age Detection Model – 2023:**
Developed Pose Estimation-based Age Recognition, analyzing key points for age prediction. The pose estimation key points helped the model attain better (slightly better) detection of the age class between adults and children (with/without faces).
- **Regression Analysis – Happiness Rate Study:**
Conducted multiple linear regression analysis in R to model global happiness rates against socioeconomic factors (GDP, personal freedom, poverty, etc.), validating model assumptions (normality, homoscedasticity, multicollinearity) and achieving 100% prediction accuracy on the test set.
- **Tesla Stock Time Series Analysis :**
Applied ARIMA modeling in R to forecast Tesla stock prices, using ACF/PACF analysis, log transformations, and ADF stationarity tests to identify the optimal ARIMA(5,2,3) model with validated residual normality and independence.
- **Klangoo Fintech Award (ZAKA) – 2022:**
Secured 3rd place in an international NLP competition for sentiment analysis in financial data. Made a system that gets twitter tweets for certain topics and creates some sentiment analysis to deduce general sentiment.
- **MNIST Digit Recognition (Without Libraries) – 2022:**
Built a digit recognition model from scratch using pure Python and NumPy, without PyTorch nor Tensorflow. Building the matrix multiplication of the learning back propagation and Gradient descent process with Python.

SCIENTIFIC PUBLICATIONS

 Lakis, R., Constantin, J., Constantin, I., Hoang, V.T., Ruichek, Y. : "Pose Estimation Keypoints in Age Recognition of Full Body Image" – 2023 IEEE IMCET  [IEEE Link](#)

 Yamout, F., Lakis, R. : "Improved TFIDF Weighting Techniques in Document Retrieval" – 2018 IEEE ICDIM  [IEEE Link](#)

LANGUAGES

- Arabic: Native || English: Fluent || French: Intermediate